

$$\begin{aligned} -0.1 \varphi_1 - 0.0667 \varphi_2 + 0.3333 \varphi_3 - 0.1667 \varphi_4 &= 4 \\ -0.125 \varphi_1 - 0.4 \varphi_2 - 0.1667 \varphi_3 + 0.6917 \varphi_4 &= -12 \end{aligned}$$

Где

$$\begin{aligned} G_1 &= \frac{1}{R_1} - \frac{1}{\infty} = 0.05 \text{ S} \\ G_2 &= \frac{1}{R_2} - \frac{1}{\infty} = 0.066667 \text{ S} \\ G_3 &= \frac{1}{R_3} - \frac{1}{\infty} = 0.1 \text{ S} \\ G_5 &= \frac{1}{R_5} - \frac{1}{6} = 0.166667 \text{ S} \\ G_6 &= \frac{1}{R_6} - \frac{1}{8} = 0.125 \text{ S} \\ G_4 &= \frac{1}{R_4} - \frac{1}{8} = 0.125 \text{ S} \end{aligned}$$

Решим систему с помощью матрицы

$$\varphi = (AG_B A^T)^{-1} (AJ_B - AG_B E_B)$$

После расчета получим

$$\begin{aligned} \varphi_1 &= 140 \text{ V} \\ \varphi_2 &= 259.064 \text{ V} \\ \varphi_3 &= 210 \text{ V} \\ \varphi_4 &= 208.374 \text{ V} \end{aligned}$$

Определим токи ветвей

$$I_1 = \frac{\varphi_1 - \varphi_2 + E_1}{R_1} = \frac{140 - (259.064) + 150}{20} = 1.547 \text{ A}$$

$$I_2 = \frac{\varphi_3 - \varphi_2 + E_2}{R_2} = \frac{210 - (259.064) + 150}{15} = 6.729 \text{ A}$$

$$I_3 = \frac{\varphi_1 - \varphi_3 + E_3}{R_3} = \frac{140 - (210) + 140}{10} = 7 \text{ A}$$

$$I_5 = \frac{\varphi_3 - \varphi_4}{R_5} = \frac{210 - (208.374)}{6} = 0.271 \text{ A}$$

$$I_0 = (\varphi_2 - \varphi_4) G_0 = (259.064 - (208.374)) \cdot 0.4 = 20.276 \text{ A}$$

$$J_0 = J_0 = 12 \text{ A}$$

$$I_6 = \frac{\varphi_1 - \varphi_4}{R_6} = \frac{140 - (208.374)}{8} = 8.547 \text{ A}$$

$$I_4 = \frac{\varphi_2 - \varphi_1 + E_4}{R_4} = \frac{0 - (140) + 140}{8} = 0 \text{ A}$$